

Assignment: AI System Specification

Course: Introduction to Artificial Intelligence

Topic: PEAS Framework and Task Environment Analysis

Application: An AI Tutor for personalized student learning recommendations.

PEAS Specification:

Performance Measures:

- Improvement in student marks
- Correctness of recommendations
- Time spent learning on system
- Completion of quizzes and tasks
- Basic understanding improvement over time

Environment:

The system works in an online learning platform where students study using mobile or web. It includes basic study material like notes, videos, and quizzes. Student behavior and internet usage also affect the system.

Actuators:

- Suggest next topic
- Show quizzes and exercises
- Give feedback on answers
- Adjust difficulty level
- Send reminders

Sensors:

- Student clicks and activity
- Quiz scores
- Time spent on topics
- Previous results
- Simple user preferences

Environment Classification:

Partially Observable: Because system cannot directly know what student actually understands.

Multi-agent: Many students use the system at the same time.

Stochastic: Student performance can change depending on different factors.

Sequential: Each step depends on previous learning.

Dynamic: Student progress keeps changing over time.

Continuous: Learning is not fixed and changes gradually.

Critical Analysis:

→ **Partial observability is a difficult part** because we cannot directly see how much a student understands. We only see their quiz results and activity, so we must estimate their knowledge.

→ **Utility Function:** A simple utility function:

$$U = \text{learning} + \text{engagement} - \text{time}$$

This means the system tries to improve learning, keep students interested, and reduce wasted time.

If learning is given more importance, the system will focus more on difficult topics, but it may reduce student interest.

Structured Representation (JSON):

```
{
  "system_name": " An AI Tutor for personalized student learning
recommendations ",
  "peas": {
    "performance_measures": [
      "Improvement in student marks",
      "Correctness of recommendations",
      "Time spent learning on system",
      "Completion of quizzes and tasks",
      "Basic understanding improvement over time"
    ],
    "environment": "The system works in an online learning platform where
students study using mobile or web. It includes basic study material like
notes, videos, and quizzes. Student behavior and internet usage also affect
the system.",
    "actuators": [
      "Suggest next topic",
      "Show quizzes and exercises",
      "Give feedback on answers",
      "Adjust difficulty level",
```

```

        "Send reminders"
    ],
    "sensors": [
        "Student clicks and activity",
        "Quiz scores",
        "Time spent on topics",
        "Previous results",
        "Simple user preferences"
    ]
},
"environment_classification": {
    "observability": {
        "choice": "Partially Observable",
        "justification": "Because system cannot directly know what student
actually understands."
    },
    "agents": {
        "choice": "Multi-agent",
        "justification": "Many students use the system at the same time."
    },
    "stochastic": {
        "choice": "Yes",
        "justification": "Student performance can change depending on
different factors."
    },
    "sequential": {
        "choice": "Yes",
        "justification": "Each step depends on previous learning."
    },
    "dynamic": {
        "choice": "Yes",
        "justification": "Student progress keeps changing over time."
    },
    "continuous": {
        "choice": "Yes",
        "justification": "Learning is not fixed and changes gradually."
    }
},
"critical_analysis": {
    "partial_observability": "Partial observability is a difficult part
because we cannot directly see how much a student understands. We only see
their quiz results and activity, so we must estimate their knowledge.",
    "utility_function_explanation": "If learning is given more importance,
the system will focus more on difficult topics, but it may reduce student
interest."
},
"utility_function": "U = learning + engagement - time"
}

```

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